

Forecasting Cryptocurrency Price Trends Using ARIMA and LSTM

Harideep Aepuri

Data Science

Regis University, Denver, CO, USA

haepuri@regis.edu

Abstract

This study compared five time series forecasting models for predicting Bitcoin (BTC/USDT) close price at one-minute intervals. One-minute open-high-low-close-volume (OHLCV) data spanning 60 days from January 30 to March 31, 2026 was collected from the Binance exchange API, producing 86,341 records. The dataset was divided into a 45-day training set and a 15-day true out-of-sample test set. Five models were implemented and evaluated under a consistent rolling one-step-ahead protocol: ARIMA(2,1,2), Facebook Prophet, LSTM Univariate, LSTM Multivariate, and a Hybrid ARIMA+LSTM model. Feature engineering produced 43 technical indicators from raw OHLCV data, and XGBoost feature importance analysis reduced the multivariate LSTM input to 20 features. Evaluation used four metrics: mean absolute error (MAE), root mean squared error (RMSE), mean absolute percentage error (MAPE), and directional accuracy. Additional analyses included walk-forward cross-validation with four expanding folds, multi-horizon forecasting at $H = 1, 5, 10, 30$ minutes, an extended 3-hour context window LSTM experiment ($W = 180$), multi-day forecast stability tests, and a cross-market study that incorporated Gold, NASDAQ, and the US Dollar Index as potential predictors for daily BTC forecasting.

ARIMA(2,1,2) has achieved the lowest error values, MAE = \$30.18 and MAPE = 0.043% on a 15-day test. Another model, Prophet has huge error values with MAE = \$8,897.93, as it requires a seasonal structure, which is absent from one-minute data. Both LSTM models underperformed ARIMA, with the univariate reaching MAE = \$64.52, and multivariate reaching MAE = \$68.91. These results are consistent with the efficient market hypothesis: Bitcoin at one-minute resolution behaves as a near-random walk, and no model achieved reliable directional prediction above chance. In the cross-market analysis, NASDAQ daily returns showed the highest correlation with BTC ($r = 0.617$), but Granger causality tests found no predictive lead-lag relationship. The multi-market daily LSTM reduced forecast error by 6.8% over a BTC-only LSTM baseline, though both remained above the naive persistence baseline on the true out-of-sample daily test.

I. INTRODUCTION AND BACKGROUND

Bitcoin is the largest cryptocurrency by market capitalization, and the BTC/USDT trading pair on the Binance exchange is among the most liquid in the world. Since its introduction in 2008 [1], Bitcoin has attracted substantial research interest in price forecasting, driven by practical applications in algorithmic trading, portfolio risk management, and exchange market design.

Classical time series forecasting relies on the autoregressive integrated moving average (ARIMA) framework. Hyndman and Khandakar [2] implemented an automatic ARIMA order selection using step-wise AIC search combined with a unit root for differencing order test, implemented a `auto_arima` in the `auto_pmdarima` library which is used in this project. ARIMA captures linear autocorrelation in stationary time series through autoregressive and moving average components combined with differencing to remove trends. The framework is computationally efficient, interpretable, and well-suited for short-horizon forecasting tasks on weakly autocorrelated series.

Long short-term memory (LSTM) networks [3] addressed the vanishing gradient problem in recurrent neural networks through gated memory cells that can retain context across long sequences. LSTM has been applied to Bitcoin price forecasting at daily resolution and reported to outperform classical time series models [4]. However, many such comparisons suffer from an evaluation inconsistency: ARIMA is evaluated using multi-step rolling forecasts while LSTM is evaluated using one-step predictions on a

static test set. This asymmetry inflates the apparent advantage of LSTM and is corrected in the present study.

This project corrected that inconsistency by implementing a rolling one-step-ahead evaluation protocol applied uniformly to all five models. Every model received identical information at prediction time: the history of prices up to step $t - 1$ was used to predict $\hat{y}_t$, and the sequence was rolled forward one step at a time across all 21,600 test records. Beyond the core comparison, the analysis was extended to multi-horizon forecasting, walk-forward cross-validation, extended context window experiments, multi-day forecast stability analysis, and a cross-market study using daily data from equity and commodity markets.

This report is organized as follows. Section II defines the research questions. Section III reviews related work. Section IV describes the methodology and model architectures. Section V presents the data analysis. Section VI reports experimental results across all analyses. Section VII presents conclusions and directions for future work.

II. PROBLEM STATEMENT

The central question this project addressed was: under a uniform rolling one-step-ahead evaluation, which forecasting method produces the lowest error for BTC/USDT close price at one-minute resolution? Three secondary research questions guided the extended analysis:

- 1) Under a consistent rolling one-step-ahead protocol, does LSTM produce lower forecast error than ARIMA?
- 2) How does forecast error grow as the prediction horizon increases from 1 minute to 30 minutes, and does the relative ranking of models change across horizons?
- 3) Do external market signals from Gold, NASDAQ, and the US Dollar Index carry useful predictive information for daily Bitcoin price forecasting?

The motivation for the first question is the evaluation inconsistency in prior work. Published results reporting LSTM superiority over ARIMA on Bitcoin data often compared multi-step ARIMA forecasts against one-step LSTM predictions, which is not a fair comparison.

The second question has direct practical significance. Trading strategies require forecasts at horizons beyond one minute. If ARIMA is more accurate at $H = 1$ but degrades faster than LSTM at $H = 10$ or $H = 30$, the optimal model choice depends on the application.

The third question is motivated by evidence that Bitcoin co-moves with equity indices and commodity markets [5], [6]. If these correlations persist at daily resolution, adding external signals to a multivariate model could reduce forecast error.

The efficient market hypothesis in its weak form [7] predicts that past prices are insufficient for forecasting future prices in an efficient market. If this hypothesis holds for Bitcoin at one-minute resolution, no model should consistently outperform a naive persistence forecast. Testing this prediction was an implicit objective of the entire study.

Figure 1 shows the train/test split used throughout the project.

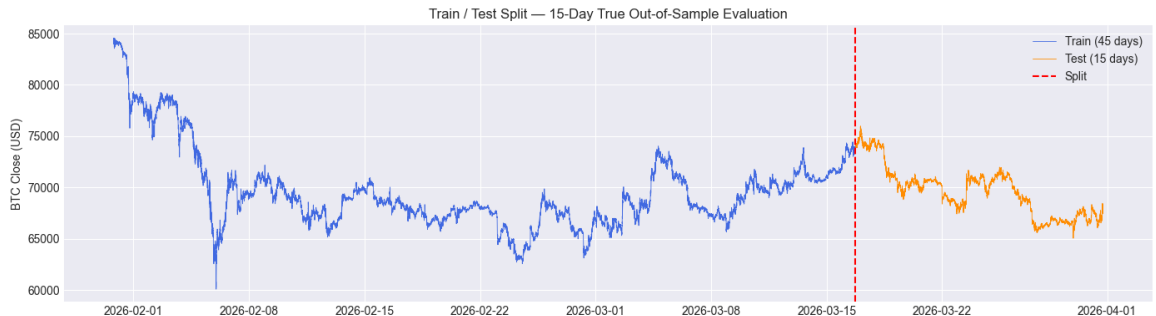


Fig. 1. Train/test split for the BTC/USDT one-minute dataset. The training set, covered from January 30 to March 16, 2026, contains 64,741 records. The test set covered March 16 to March 31, 2026 (21,600 records).

III. RELATED WORK

Hyndman and Khandakar [2] introduced the `auto_arima` algorithm, which automates ARIMA model selection through a stepwise AIC search over candidate (p, d, q) orders, using unit root tests to determine the required differencing order. Their methodology is implemented in the `pmdarima` Python library and was applied directly in this project to select ARIMA(2,1,2). ARIMA remains widely used in financial forecasting because it is computationally inexpensive, interpretable, and often competitive with more complex models on short-horizon tasks. For near-random-walk series, ARIMA(0,1,0) or low-order specifications are frequently optimal.

The LSTM Networks are introduced by Hochreiter and Schmidhuber [3] which are used for resolving the vanishing gradient problem in vanilla recurrent neural networks. LSTM cells contain input, forget, and output gates that regulate how information flows and persists across time steps. This architecture became the standard recurrent network for sequence modeling and has been applied to many financial forecasting tasks.

McNally, Roche, and Caton [4] applied a Bayesian-optimised recurrent neural network and LSTM to Bitcoin daily closing price prediction, reporting 52% directional accuracy for LSTM and concluding that deep learning outperformed ARIMA, which they described as performing poorly. However, their evaluation did not apply a consistent rolling one-step-ahead protocol to both model classes simultaneously. The present study corrects this by applying the same rolling evaluation to all five models. At one-minute resolution with a uniform evaluation framework, ARIMA substantially outperforms LSTM, showing that time resolution and evaluation fairness are decisive factors in the ARIMA versus LSTM comparison.

Taylor and Letham [8] introduced Prophet as a scalable decomposable forecasting model for business time series with strong trend and seasonal patterns. Prophet fits trend, seasonality, and holiday effects separately using Fourier series. The model performs well on daily and weekly business metrics but requires identifiable periodic structure. At one-minute resolution, Bitcoin prices do not exhibit stable seasonality, which explains Prophet's failure in this study.

Adebiyi, Adewumi, and Ayo [9] compared ARIMA and artificial neural networks for NYSE stock price prediction and found that neural networks outperformed ARIMA on their datasets. Their result reflects a common pattern in daily-frequency financial forecasting, where the additional capacity of neural networks provides some benefit. The present study finds the reverse at one-minute cryptocurrency resolution: the linear ARIMA model outperforms LSTM variants, consistent with the near-random-walk character of high-frequency price data where nonlinear patterns are absent.

Fama [7] established an efficient market hypothesis framework, which states that current prices are always reflect the past prices information. Prior empirical work has found that cryptocurrency markets show some inefficiency relative to traditional equity markets, particularly at lower frequencies

[6], The results of this study suggest that at one-minute resolution, Bitcoin behaves in a manner consistent with the weak form of the hypothesis.

Dyhrberg [5] examined the relationship between Bitcoin, Gold, and the US dollar using GARCH volatility models and found that Bitcoin shares hedging characteristics with Gold. Ji et al. [6] studied The time-varying connections are studied across cryptocurrency markets and traditional assets to find time-varying correlations with equity indices. Both studies motivated the cross-market analysis conducted in Section VI.

Chen and Guestrin [10] introduced XGBoost as a scalable and regularized gradient-boosted tree algorithm. In this project, XGBoost was used for feature importance analysis to rank the 43 engineered indicators and select the top 20 as multivariate LSTM inputs.

IV. METHODOLOGY

A. Data Collection

BTC/USDT one-minute OHLCV data was collected from the Binance exchange API by using the Python libraries(`python-binance` client library). The collected data is from January 30, 2026, to March

31, 2026, producing 86,341 records. Each record contains the open, high, low, and close prices and the trade volume for one minute. Sparse gaps caused by exchange downtime were forward-filled to maintain a continuous time index. BTC data was stored as a CSV file, and there are no external preprocessing services used.

B. Feature Engineering

Forty-three features were derived from the raw OHLCV data, organized into five groups. Technical indicators included the Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD) with its signal line, Bollinger Band upper and lower bounds, Average True Range (ATR), On-Balance Volume (OBV), and Volume-Weighted Average Price (VWAP). Lag features provided the close price at lags 1 through 7. Mean and standard deviation were captured by rolling statistics of close price for 5 minute, 15-minute and, 60-minute windows. Return features included the one-period log return and the intraday return. Cyclical time features encoded hour-of-day and day-of-week as sine and cosine transforms to preserve circular continuity across period boundaries.

XGBoost was trained on the full 43-feature set with close price as the prediction target. The top 20 features by importance score were selected for the LSTM Multivariate model. Figure 2 shows the ranked importance scores.

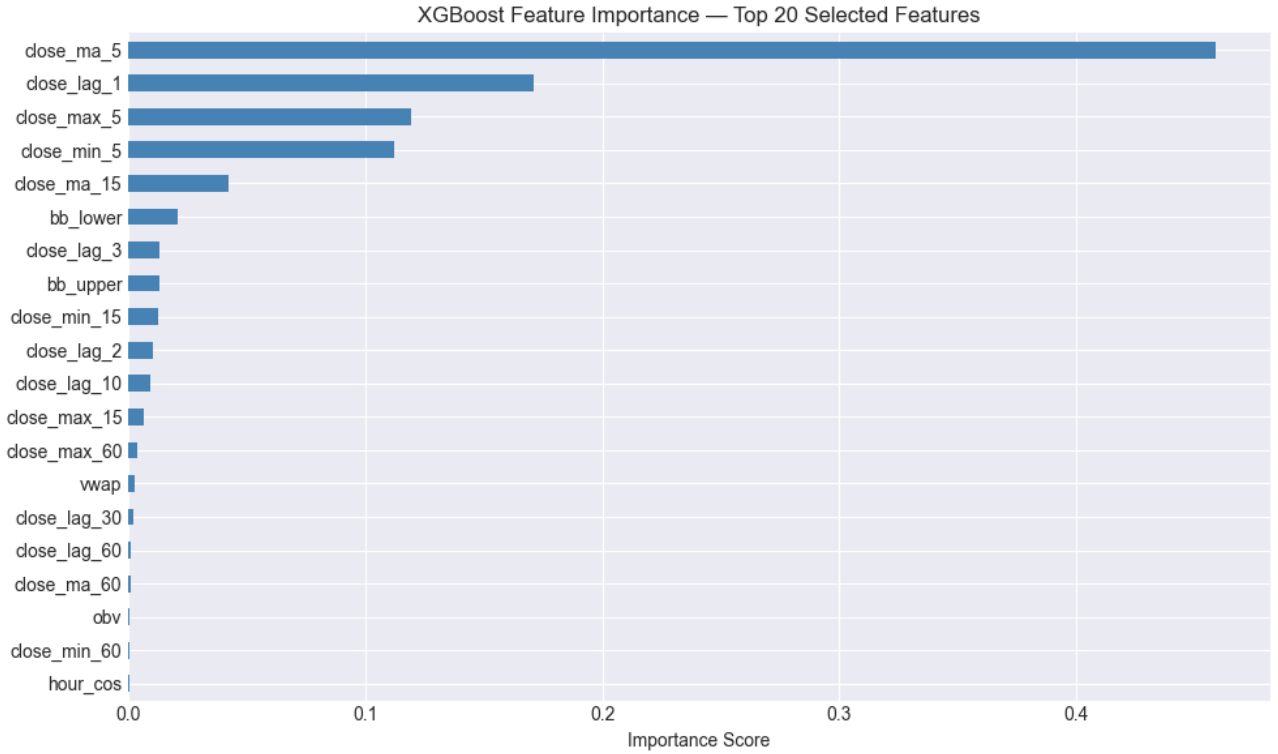


Fig. 2. XGBoost feature importance scores for all 43 engineered features. The top 20 features by score were used as inputs to the LSTM Multivariate model.

C. Evaluation Protocol

All five models were evaluated using a rolling one-step-ahead protocol. At each time step t in the test set, the model predicted $\hat{y}_t$ from information available through $t - 1$, and the prediction was compared to the actual close price y_t . Four metrics were computed over the full 15-day (21,600-step) test:

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t| \quad (1)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2} \quad (2)$$

$$\text{MAPE} = \frac{100\%}{n} \sum_{t=1}^n \frac{|y_t - \hat{y}_t|}{y_t} \quad (3)$$

$$\text{Dir. Acc.} = \frac{1}{n} \sum_{t=1}^n \mathbf{1}[\text{sign}(\hat{y}_t - y_{t-1}) = \text{sign}(y_t - y_{t-1})] \quad (4)$$

The test window for each Fold is 10% of full dataset which is $86,341 * 0.10 = 8,634$ that is equal to six days of one-minute data. Each fold training window is expanded by 8,634 records before the previous fold. The initial training window spanned 51,805 records (approximately 36 days); by Fold 4 it had grown to 77,707 records (approximately 54 days).

D. Model Architectures

Table I summarizes the architecture and key hyperparameters for each model.

TABLE I
MODEL ARCHITECTURES AND KEY HYPERPARAMETERS

Model	Architecture	Key Parameters
ARIMA(2,1,2)	Autoregressive	$p = 2, d = 1, q = 2$; order selected by AIC via <code>pmdarima</code>
Prophet	Decomposable additive	Daily and weekly seasonality; change-point prior scale 0.05
LSTM Univariate	2-layer stacked LSTM	100 and 50 units, $W = 60$, Adam lr 0.001, 25 epochs, batch 128
LSTM Multivariate	2-layer stacked LSTM	100 and 50 units, $W = 60$, 20 input features, same training config
Hybrid ARIMA+LSTM	Two-stage	ARIMA(2,1,2) base; LSTM (64 and 32 units) trained on 3σ -clipped residuals

ARIMA: The model order was selected by `pmdarima`'s `auto_arima` using AIC on the training set. Rolling one-step evaluation used `statsmodels`'s `apply()` method to update fitted values at each step without full model refit, giving $O(n)$ evaluation cost over the 21,600-step test.

LSTM Univariate: The model used a 60-step lookback window and was trained on the scaled close price only. Input was scaled to $[0, 1]$ using a `MinMaxScaler` fit on training data alone. Two stacked LSTM layers (100 and 50 units, tanh activation) fed a `Dense(1)` output layer.

LSTM Multivariate: The Multivariate matched the univariate version, but this model used the top 20 features selected by XGBoost as input. Two separate `MinMaxScaler` instances were fit on training data one for the 20 input features and one for the close price target. Inverse transformation of predictions used only the target scaler, preventing data leakage.

Hybrid ARIMA+LSTM: ARIMA(2,1,2) was fit on the training set and in-sample residuals were extracted. Residuals were clipped at $\pm 3\sigma$ to remove a single initialization outlier at step 0. A smaller LSTM (64 and 32 units, $W = 60$) was trained on the clipped residuals. Each test step's prediction was

the ARIMA one-step forecast plus the LSTM residual correction, updated with the observed residual at each step.

V. DATA ANALYSIS

A. Dataset Overview

The dataset contained 86,341 one-minute records spanning January 30 to March 31, 2026. The close price series showed a moderate trend over the 60-day period with episodes of elevated volatility. Due to high-activity market sessions, the volumes are irregular with periodic spikes. The autocorrelation function showed significant autocorrelation at the first lag and rapid decay thereafter, consistent with a near-random walk process where the most recent value is the best predictor of the next. Figure 3 shows the exploratory data analysis overview: the full price history, one-minute return distribution, volume profile, and autocorrelation function.



Fig. 3. Exploratory data analysis of the 86,341-record BTC/USDT dataset. Top left: close price over 60 days. Top right: distribution of one-minute log returns. Bottom left: trade volume profile. Bottom right: autocorrelation function of close price.

B. Stationarity Analysis

The Augmented Dickey-Fuller (ADF) test on the raw close price series failed to reject the null hypothesis of a unit root at the 5% significance level, confirming that the series was non-stationary in levels. First differencing produced a stationary series, which established $d = 1$ for the ARIMA specification. The Partial autocorrelation function on the differenced series it showed a partial autocorrelation at lags 1 and 2 which results $p = 2$ and the autocorrelation function decay pattern supported $q = 2$ by the AIC optimal order found by `auto_arima`.

C. Train/Test Split and Data Quality

The data was split into training and test sets of 64,741 from January 30, 2026, to March 16, 2026, and 21,600 as a test set from March 16, 2026, to March 31, 2026, that represents the 45 training days and 15 test days, respectively. All scalers, feature pipelines, and model parameters were derived exclusively from training data. The test set was held out until the final evaluation to ensure a true out-of-sample performance estimate. A small number of missing records from exchange downtime were identified through timestamp gap analysis and addressed by forward-filling the preceding record's values.

D. Cross-Market Data Sources

For the cross-market analysis, three external daily price series were collected alongside the BTC/USDT data. Gold features(ticker GC=f),NASDAQ composite Index (ticker ^IXIC), and US Dollar index (ticker: DX-Y.NYB), all the data collected from `yfinance` python library from January 30, 2026, to March 31, 2026. Each series provided daily closing prices for the same 60-day window as the BTC dataset.

BTC daily close prices were derived by resampling the one-minute series to a daily frequency, taking the last available close price of each calendar day. All four daily series (BTC, Gold, NASDAQ, DXY) were aligned on the intersection of their trading days, as Gold futures and the DXY trade on different schedules from US equity indices. Daily log returns were computed for all series before correlation and Granger causality analyses to ensure stationarity.

No survivorship bias or look-ahead leakage affected the cross-market data, as all series were downloaded in their raw historical form. The daily LSTM models trained on this data used a chronological train/test split consistent with the one-minute analysis: models were trained on the first 45 days and tested on the final 15 days.

VI. RESULTS

A. One-Step-Ahead Forecast Comparison

Table II presents the performance of all five models on the 15-day test set.

TABLE II
MODEL PERFORMANCE ON THE 15-DAY OUT-OF-SAMPLE TEST (ROLLING ONE-STEP-AHEAD)

Model	MAE (\$)	RMSE (\$)	MAPE (%)	Dir. Acc. (%)
ARIMA(2,1,2)	30.18	46.46	0.0433	49.6
Prophet	8897.93	10155.09	13.0135	50.1
LSTM Univariate	64.52	96.26	0.0928	48.3
LSTM Multivariate	68.91	103.54	0.0990	48.8
Hybrid ARIMA+LSTM	30.21	46.48	0.0433	49.6

ARIMA(2,1,2) produced the lowest MAE and RMSE, followed closely by the Hybrid model. The difference between the two was negligible: MAE of \$30.18 vs.\$30.21, and RMSE of \$46.46 vs.\$46.48. This near-identical performance confirmed that the LSTM component of the Hybrid model did not learn additional meaningful patterns from the ARIMA residuals, which behaved effectively white noise,only very limited additional non-linear patterns are left for LSTM component to learn.

Prophet produced an MAE of \$8,897.93, roughly 295 times larger than ARIMA. Prophet’s decomposable model assumes identifiable trend and seasonality components; neither is present in one-minute Bitcoin data. The model’s forecasts diverged systematically from the actual price series across the 15-day test.

Both LSTM variants produced errors approximately twice as large as ARIMA. The multivariate LSTM (MAE = \$68.91) was slightly worse than the univariate LSTM (MAE = \$64.52), indicating that the additional 20 engineered features added noise rather than predictive signal at one-minute resolution. All five models produced directional accuracy values between 48.3% and 50.1%, consistent with random price movements where no model reliably predicts the direction of the next one-minute change.

Figure 4 presents all four metrics visually.

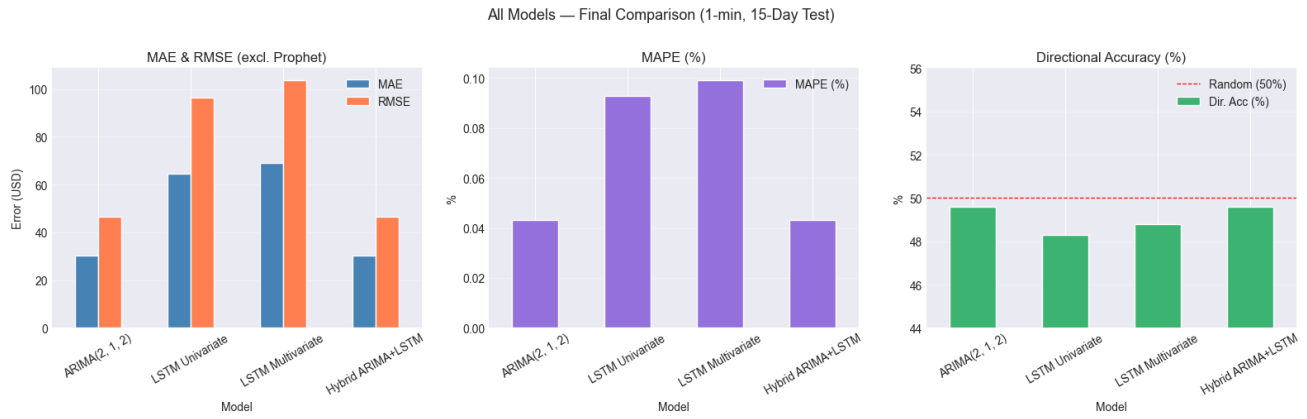


Fig. 4. Four-metric comparison across all five models on the 15-day out-of-sample test. Prophet is excluded from the MAE and RMSE bar charts for scale clarity.

B. Individual Model Forecasts

Figure 5 shows rolling one-step-ahead forecasts against actual BTC close price for the final 48 hours of the test period for four of the five models. Figure 6 shows the Prophet forecast separately due to the scale difference.

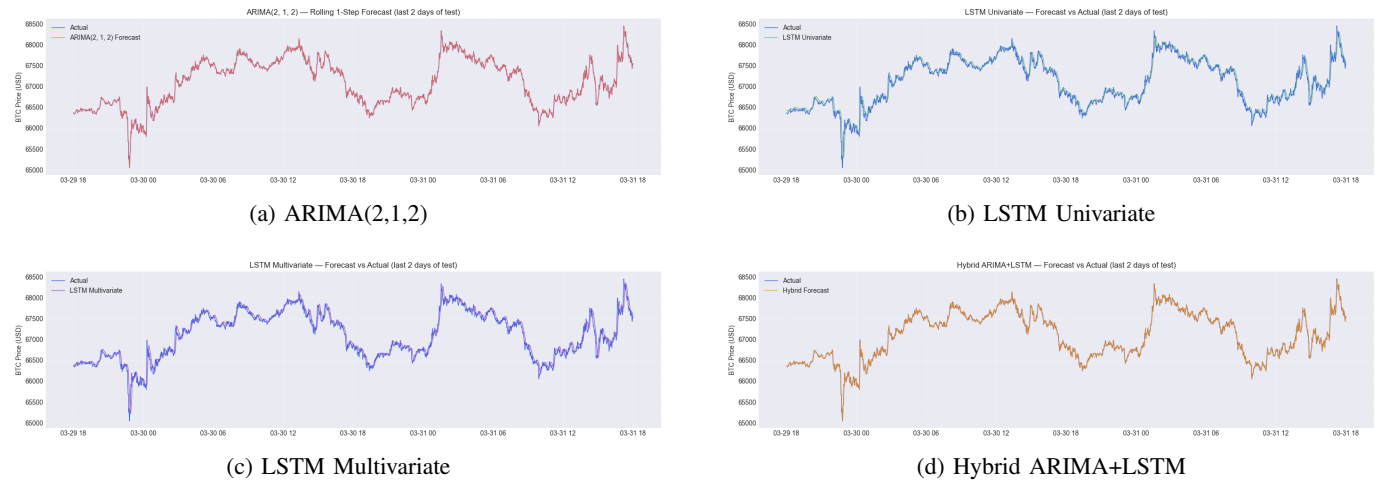


Fig. 5. Rolling one-step-ahead forecasts vs. actual BTC/USDT close price for the final 48 hours of the test set. ARIMA (a) and Hybrid (d) track the actual series closely. Both LSTM models (b, c) show a persistent lag.

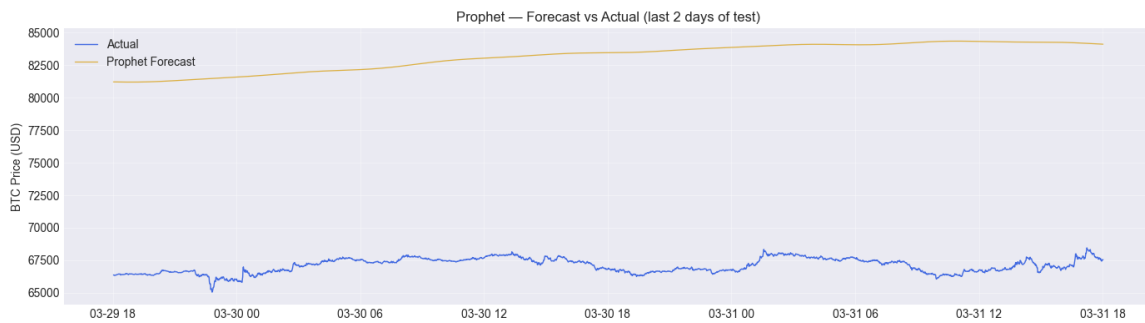


Fig. 6. Prophet forecast vs. actual BTC/USDT close price. The model's forecast diverges from the actual series within the first hours of the test period due to the absence of stable seasonal structure in one-minute financial data.

C. Walk-Forward Cross-Validation

Walk-forward cross-validation with four expanding folds was applied to ARIMA to measure performance stability as the training window grows. Table III reports fold-level MAE values.

TABLE III
ARIMA WALK-FORWARD CROSS-VALIDATION (4 FOLDS)

Fold	MAE (\$)
Fold 1	37.28
Fold 2	31.24
Fold 3	30.75
Fold 4	26.80
Mean $\pm$ Std	31.52 $\pm$ 4.33

MAE declined consistently from Fold 1 to Fold 4, showing that additional training data produced modest but steady improvement in ARIMA’s forecast accuracy. The standard deviation of \$4.33 was small relative to the mean, confirming stable generalization. The cross-validation mean MAE of \$31.52 was consistent with the final 15-day test MAE of \$30.18, validating that the test result was not an artifact of the specific evaluation period.

D. Multi-Horizon Forecast Analysis

ARIMA and LSTM Univariate were evaluated at $H = 1, 5, 10$, and 30 minutes using recursive multi-step forecasting to assess how accuracy degrades with prediction horizon. Table IV reports the MAE at each horizon.

TABLE IV
MULTI-HORIZON MAE (\$) FOR ARIMA AND LSTM UNIVARIATE

Model	$H = 1$	$H = 5$	$H = 10$	$H = 30$
ARIMA(2,1,2)	32.96	72.76	100.35	166.10
LSTM Univariate	69.36	87.18	126.28	215.81

Both models showed monotonically increasing MAE with horizon, which is expected for a near-random-walk series. ARIMA maintained a consistent advantage over LSTM at all horizons. At $H = 1$, ARIMA’s MAE was approximately 48% of LSTM’s; at $H = 30$, it was approximately 77% of LSTM’s, indicating that the absolute gap widened but the relative advantage compressed at longer horizons. Neither model was sufficiently accurate for practical trading at the 30-minute horizon.

Figure 7 illustrates the MAE and RMSE growth curves.

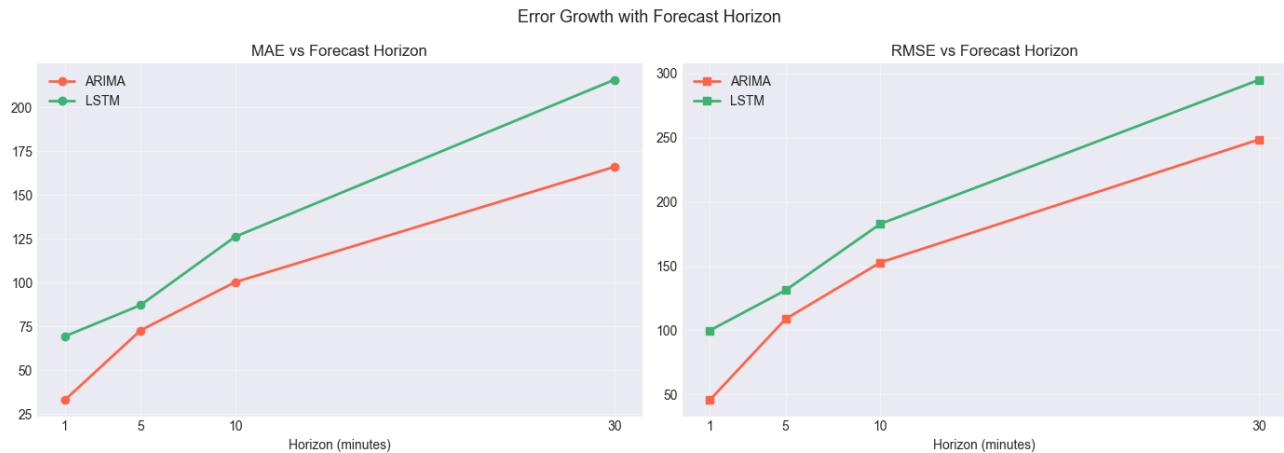


Fig. 7. MAE and RMSE growth as prediction horizon increases from 1 to 30 minutes for ARIMA and LSTM Univariate. Both metrics increase monotonically; ARIMA maintains lower error across all horizons.

E. Extended Context Window ($W = 180$)

A separate LSTM was trained with a 3-hour (180-step) lookback window to test whether longer temporal context improves forecast quality. Table V reports multi-horizon results for this model.

TABLE V
LSTM WITH $W = 180$ (3-HOUR CONTEXT): MULTI-HORIZON RESULTS

Horizon	MAE (\$)	Directional Acc. (%)
$H = 1$ min	53.61	52.7
$H = 5$ min	72.02	54.3
$H = 10$ min	99.82	51.0
$H = 30$ min	168.24	50.7

The 3-hour context window LSTM achieved 54.3% directional accuracy at $H = 5$ minutes, the highest directional accuracy observed across all models and settings in this study. This was the only experimental configuration where a model showed a consistent edge above 50%. The improvement was modest but repeatable, suggesting that 3-hour price history encodes some medium-frequency signal at the 5-minute forecast horizon that a 60-step window cannot capture.

Figure 8 shows MAE and directional accuracy across horizons for the $W = 180$ model.



Fig. 8. LSTM with 3-hour context window ($W = 180$): MAE and directional accuracy as a function of prediction horizon. Directional accuracy peaks at 54.3% for $H = 5$ minutes.

F. Multi-Day Forecast Stability

Each model was evaluated over 2-day, 5-day, 10-day, and 15-day subsets of the test set to assess whether performance was stable across evaluation window lengths. Table VI reports MAE for each combination.

TABLE VI
MULTI-DAY FORECAST STABILITY: MAE (\$) BY EVALUATION WINDOW

Model	2-Day	5-Day	10-Day	15-Day
ARIMA(2,1,2)	35.65	32.12	31.90	30.18
Prophet	893.69	3427.51	6044.10	8897.93
LSTM Univariate	70.80	63.76	64.22	64.52
LSTM Multivariate	78.80	71.19	71.14	68.91
Hybrid ARIMA+LSTM	35.66	32.15	31.93	30.21

ARIMA and Hybrid MAE decreased steadily as the evaluation window lengthened, reflecting the averaging benefit of longer evaluation periods on near-zero-mean residuals. Both LSTM univariate and LSTM Multivariate MAE values shows a stable window length ranging from 63–79, which indicates a limited sensitivity to the evaluation period, and Prophet error grew rapidly with window length, confirming systematic drift that compounded over time. Figure 9 shows these trajectories.



Fig. 9. MAE as a function of evaluation window length for each model (Prophet excluded for scale). ARIMA and Hybrid error decreases with longer windows; LSTM error stabilizes.

G. Cross-Market Analysis

Daily BTC/USDT close prices were obtained by resampling the one-minute data to daily frequency. Daily closing prices for Gold (GC=F), NASDAQ (IXIC), and the US Dollar Index (DXY) were downloaded through the `yfinance` library and aligned to the 60-day BTC series.

1) *Correlation Analysis*: Daily return correlations between BTC and the three external markets were computed over the full 60-day period.

NASDAQ achieved the strong positive correlation $r = 0.617$ with BTC returns, showing that broad equity market risk-on and risk-of movements are reflected in bitcoin prices on the same day. Gold showed a weak positive correlation ($r = 0.076$), and DXY showed a weak negative correlation ($r = -0.127$), consistent with Bitcoin’s partial status as a dollar-denominated store of value.

2) *Granger Causality*: Granger causality tests were conducted to determine whether any external market’s daily returns could predict BTC daily returns at a one-day lag. Table VII reports p-values for each predictor at lags 1 through 3.

NASDAQ had the highest correlation with BTC but showed no Granger causality at any lag, which means NASDAQ and BTC move together on the same day rather than NASDAQ leading BTC by one or more days. Gold showed marginal significance at lag 3 ($p = 0.017$), but isolated significance at a single lag is unlikely to represent a robust predictive relationship, particularly given the small 60-day sample. DXY showed no predictive relationship.

3) *Daily LSTM: BTC-Only vs. Multi-Market*: Two daily LSTM models were trained to test whether adding external market signals improves daily BTC close price forecasting. The BTC-only model used

TABLE VII
GRANGER CAUSALITY TEST P-VALUES (BTC DAILY RETURNS AS DEPENDENT VARIABLE)

Predictor	Lag 1	Lag 2	Lag 3	Result
Gold	0.208	0.105	0.017	Marginally significant at lag 3 only
NASDAQ	0.264	0.406	0.310	Not significant
DXY	0.771	0.287	0.296	Not significant

past daily close prices. The multi-market model used BTC, Gold, NASDAQ, and DXY as four concurrent input features. Table VIII summarizes the results on the out-of-sample daily test.

TABLE VIII
DAILY LSTM FORECAST RESULTS (OUT-OF-SAMPLE TEST)

Model	MAE (\$)
Naive persistence baseline	1867.11
LSTM BTC-only	2288.12
LSTM Multi-market	2132.31

The multi-market LSTM analysis improved the BTC prediction by reducing 6.8% MAE compared with the BTC-Only LSTM model which results the external market variables like Gold, NASDAQ and The U.S The Dollar Index has added some useful information for forecasting daily bitcoin prices. However, both LSTM variants produced higher error than the naive persistence baseline of \$1,867.11. The result suggests that 60 days of daily observations is insufficient for LSTM to generalize well enough to beat a trivial forecast, regardless of whether external features are included.

Figure 10 shows normalized price trends and daily return correlations alongside the daily LSTM forecast comparison.

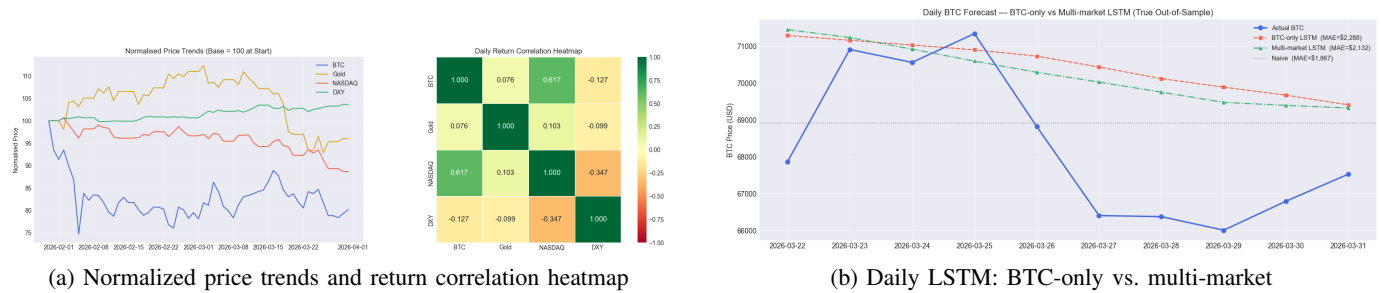


Fig. 10. Cross-market analysis. Left: co-movement of BTC, NASDAQ, Gold, and DXY over 60 days with return correlation heatmap. Right: daily BTC-only and multi-market LSTM forecasts vs. actual prices on the out-of-sample test.

VII. CONCLUSION

This project compared five forecasting models for Bitcoin price prediction at one-minute resolution under a consistent rolling one-step-ahead evaluation. ARIMA(2,1,2) produced the best results on the 15-day out-of-sample test with MAE = \$30.18, RMSE = \$46.46, and MAPE = 0.043%. LSTM models produced errors roughly twice as large, and the Hybrid ARIMA+LSTM model matched ARIMA almost exactly, confirming that ARIMA residuals at this scale contain no learnable signal for a secondary model.

The findings support the weak form of the efficient market hypothesis for Bitcoin at one-minute resolution. All five models produced directional accuracy values near 50%, and no model produced MAPE values greater than 0.1%, except for the Prophet model, which reflects the small relative magnitude of

individual one-minute price movements rather than model precision. ARIMA’s advantage over LSTM can be attributed to the absence of nonlinear structure at this frequency: a near-random-walk series is well-served by a linear autoregressive model, and the additional capacity of LSTM confers no benefit.

Multi-horizon analysis confirmed that ARIMA’s advantage holds at all tested horizons from 1 to 30 minutes. The one exception was the 3-hour context window LSTM, which achieved 54.3% directional accuracy at $H = 5$ minutes, the only configuration in the study where a model showed a consistent edge above random. This result suggests that medium-frequency patterns at the 5-minute horizon may be partially captured by a longer lookback window, though the practical significance of a 4.3 percentage point improvement over random remains limited.

Cross-Market analysis suggests that the NASDAQ daily returns show a strong positive correlation ($r = 0.617$) with BTC, that reflecting the co-movement within the same training day rather than a predictive lead-lag relationship. Granger causality tests found on significant predictive relationship for NASDAQ or DXY. The multi-market LSTM reduced daily forecast error by 6.8% relative to a BTC-only baseline, but both models fell short of the naive persistence baseline given the limited 60-day daily training set.

Future work could explore several directions. Order book data at tick resolution may carry information not present in one-minute OHLCV data and could improve short-horizon directional prediction. Regime-switching models that adapt to structural breaks in volatility may better handle Bitcoin’s non-stationary variance. On-chain metrics such as transaction volume, active addresses, and miner revenue have been proposed as leading indicators of Bitcoin price movements and represent a natural extension of the cross-market framework developed here.

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